UNIVERSITY OF BOLOGNA

Computational Imaging — 2025/26

LIMITED-ANGLE CT RECONSTRUCTION

https://github.com/Manieri03/limited-angle-ct-reconstruction

Four families of methods compared: variational, end-to-end, generative, hybrid

Group D

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01 — INTRODUCTION

The problem: limited-angle CT

A CT scan reconstructs the image *x* from projections (sinogram) *y* collected along different angles:

$$y^\delta = K x + e$$

- K = projection operator (discrete Radon transform, parallel-beam geometry)
- e = acquisition noise, additive Gaussian proportional to the norm of the sinogram
- In the limited-angle case the projections cover only a restricted subset of angles, not the full range [0°, 180°)
- A strongly ill-posed problem

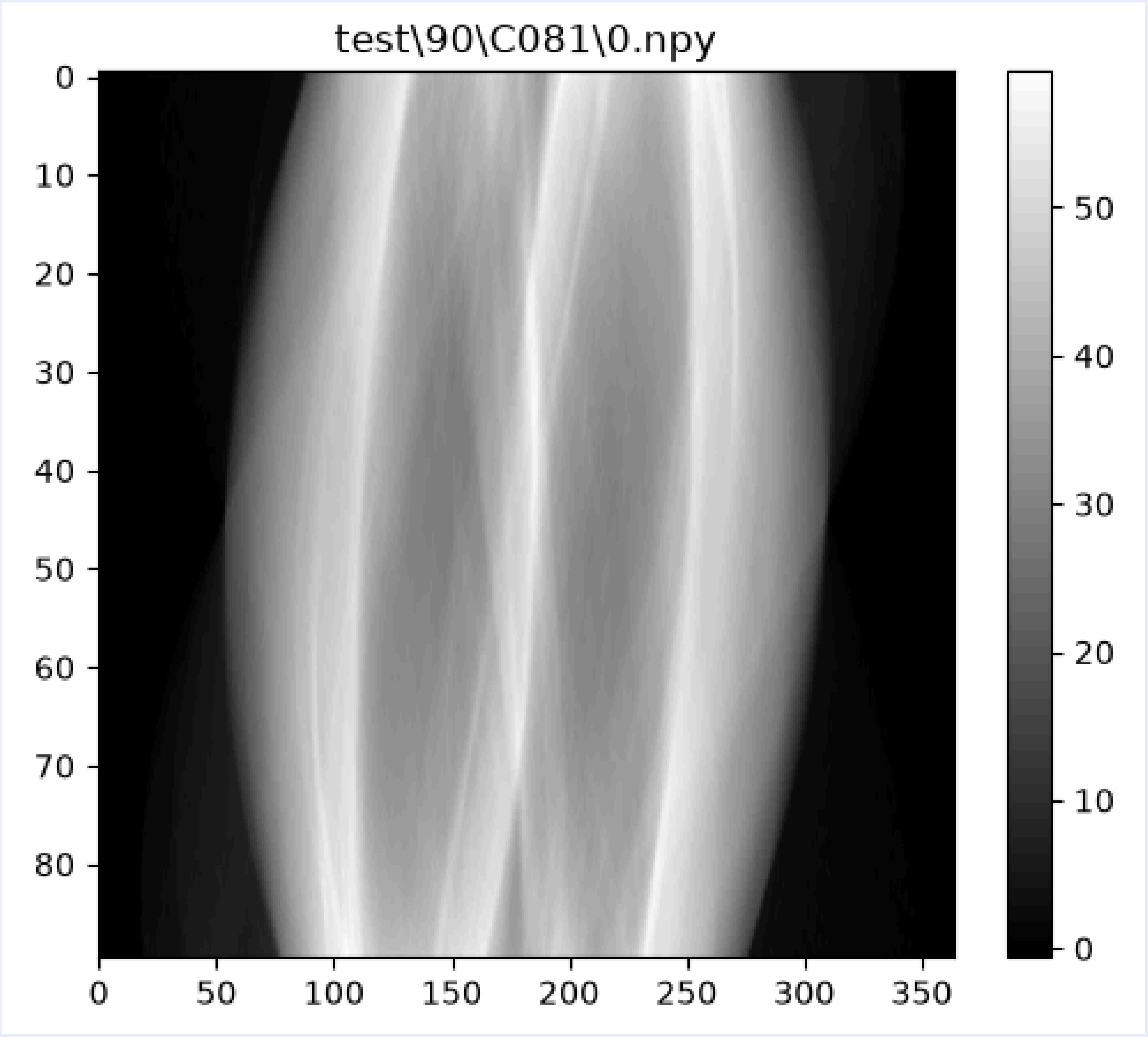


Fig. 1 — Noisy sinogram (90-angle config.): vertical axis = projection angle, horizontal axis = position on the detector

Dataset and angular configurations

Mayo dataset (low-dose CT), 256×256 images

Split by patient:	8 train / 2 validation / 1 test
Noise:	additive Gaussian on the sinogram, relative level 0.5%
Geometry:	parallel-beam, operator K via IPPy / ASTRA Toolbox
Preprocessing:	grayscale, resize to 256×256, original split
Sinogram creation:	limited-angle generation from the resized dataset

4 angular configurations

90 angles [-45°, 45°]	**45 angles** [-45°, 45°]
30 angles (-30°, 30°)	**15 angles** (-30°, 30°)

A distinct ASTRA projector is instantiated for each configuration, shared across all images of that configuration.

OVERVIEW

Four families of methods

Same test set, same degraded sinograms, same metrics (PSNR / SSIM vs ground truth)

VARIATIONAL

Total Variation

TV minimization constrained by fidelity to the sinogram, solved with Chambolle-Pock

END-TO-END

UNet

Residual encoder-decoder network mapping FBP → reconstruction, trained to imitate TV

GENERATIVE

GAN + DPS

DCGAN generator as prior, latent code optimized with data guidance (Diffusion Posterior Sampling)

HYBRID

Weighted TV

TV with a per-pixel regularization weight, estimated once from the UNet output and kept fixed

Total Variation

VARIATIONAL

Formulation (Morozov discrepancy principle):

$$x^* \in \arg\min TV(x) \quad \text{s.t.} \quad \| Kx - y\delta \|_2 \leq \varepsilon$$

with ε = NOISE_LEVEL · $\| y\delta \|_2$, consistent with the noise level used at generation time.

Algorithm: Chambolle-Pock

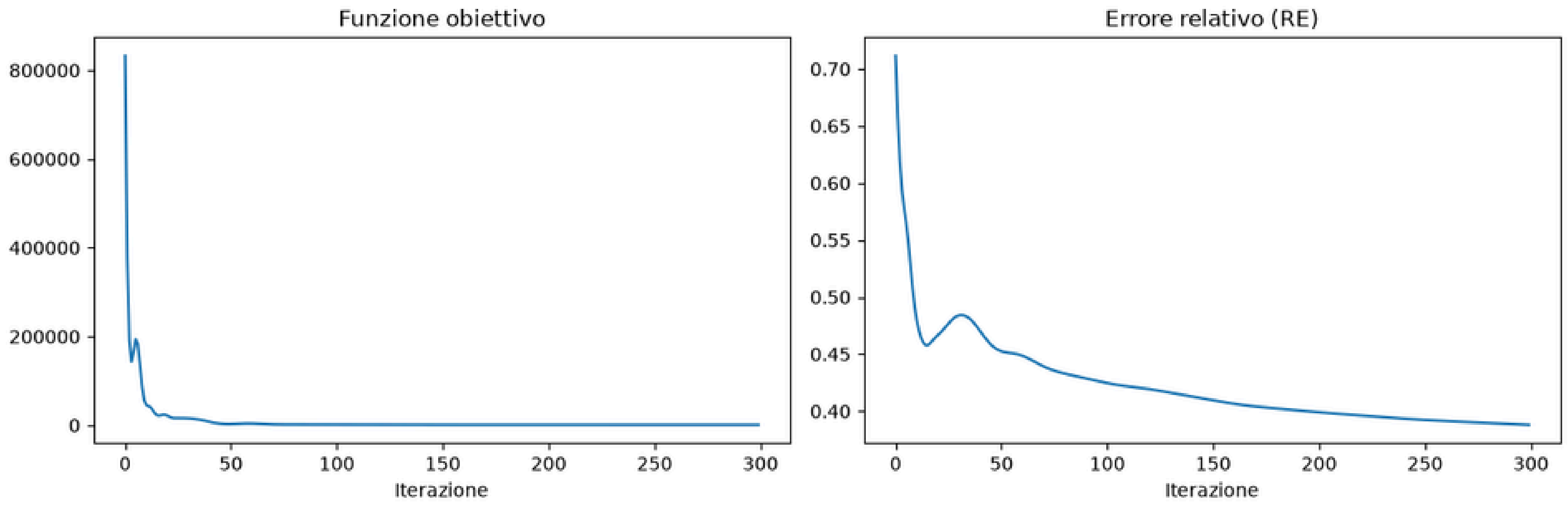


Fig. 6 — Evolution of the objective function and the relative error (RE) during the iterations of the Chambolle-Pock solver

Total Variation — tuning and result

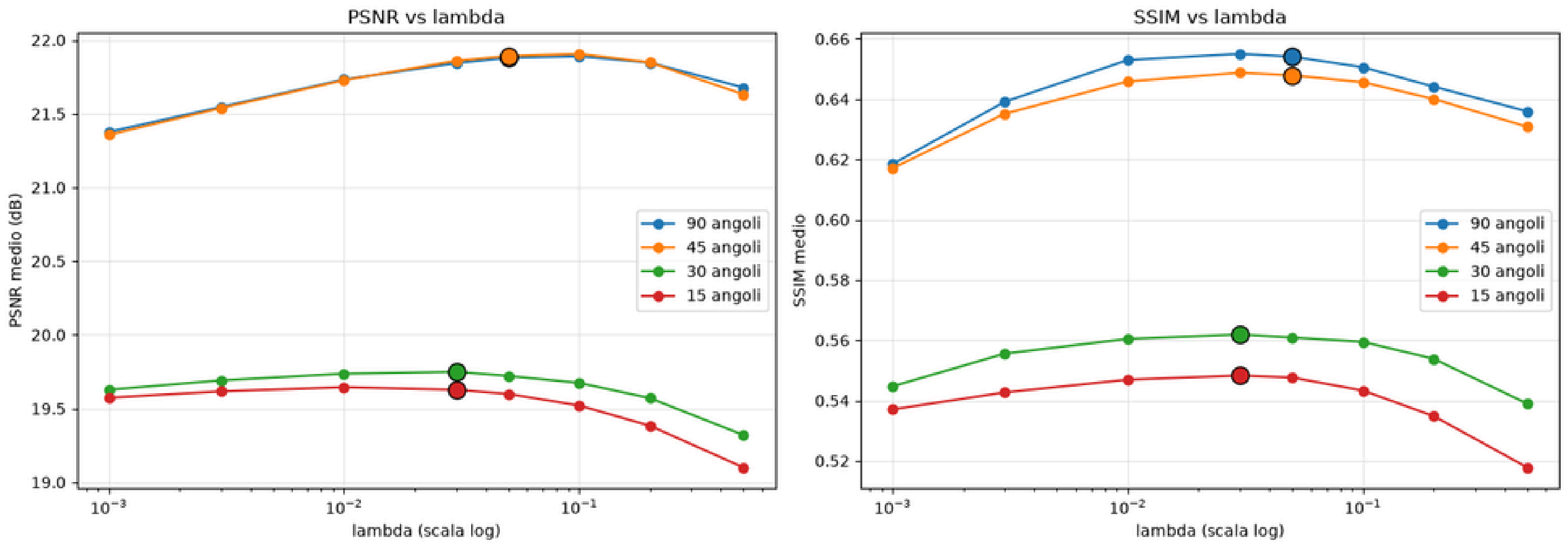


Fig. 3 — Mean PSNR/SSIM as λ varies (validation), per angular configuration

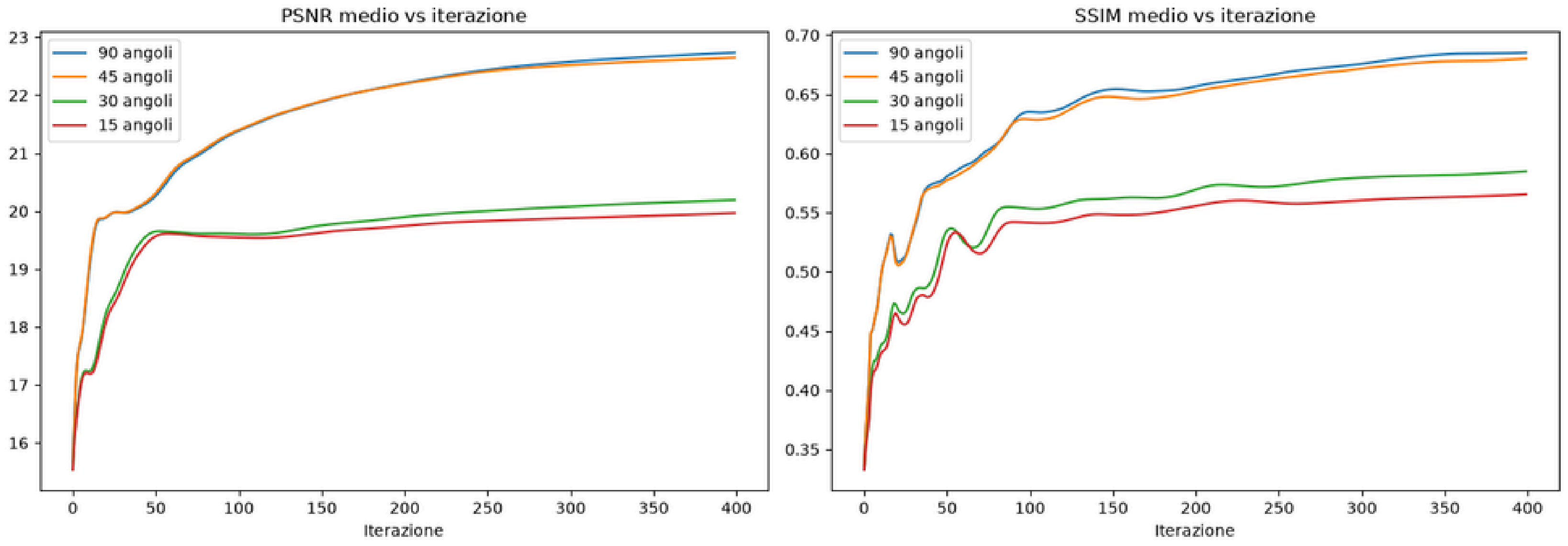


Fig. 4 — Convergence curves as the number of iterations varies: no clear plateau within the 400 tested iterations

Configuration	λ_TV	maxiter
90 angles	0.05	300
45 angles	0.05	300
30 angles	0.03	300
15 angles	0.03	300

Total Variation — example and mean metrics

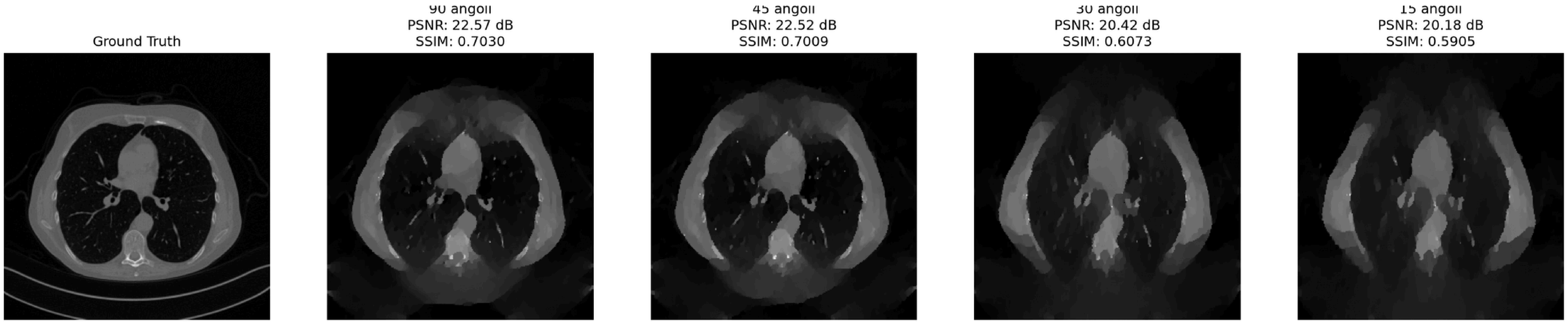
90 angles
23.83 dB ± 2.12
SSIM **0.7257**

45 angles
23.79 dB ± 2.14
SSIM **0.7221**

30 angles
21.75 dB ± 2.35
SSIM **0.6426**

15 angles
21.51 dB ± 2.37
SSIM **0.6231**

Mean PSNR and SSIM (± std. dev.) over the entire test set (327 slices), TV reconstruction only



Qualitative example (slice C081): ground truth vs TV reconstruction across the 4 angular configurations, with PSNR/SSIM for the single case

UNet — architecture

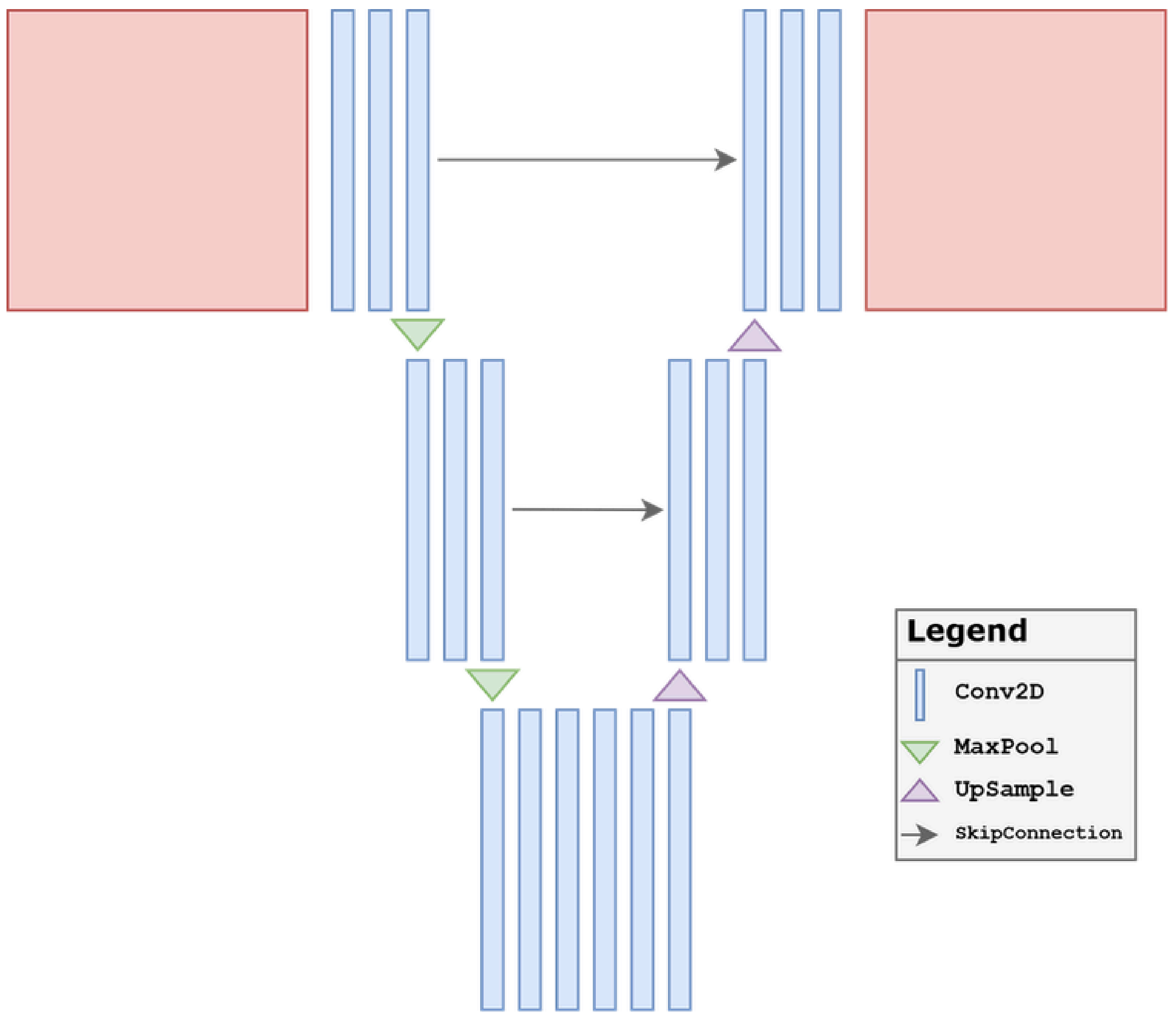
END-TO-END

Input: FBP reconstruction
Target: TV reconstruction (not the ground truth)

- Residual encoder-decoder, channels [32, 64, 128, 256], 3 down/up levels + bottleneck
- ResDownBlock / ResUpBlock blocks, 2 conv layers per block, skip connections
- Final sigmoid activation → output constrained to [0, 1]
- A separate UNet for each of the 4 angular configurations

Loss = MSE + 0.1·SSIM · Adam lr=1e-4 · 25 epochs · batch 16

FBP → Encoder → Bottleneck → Decoder → Output



UNet — training and model selection

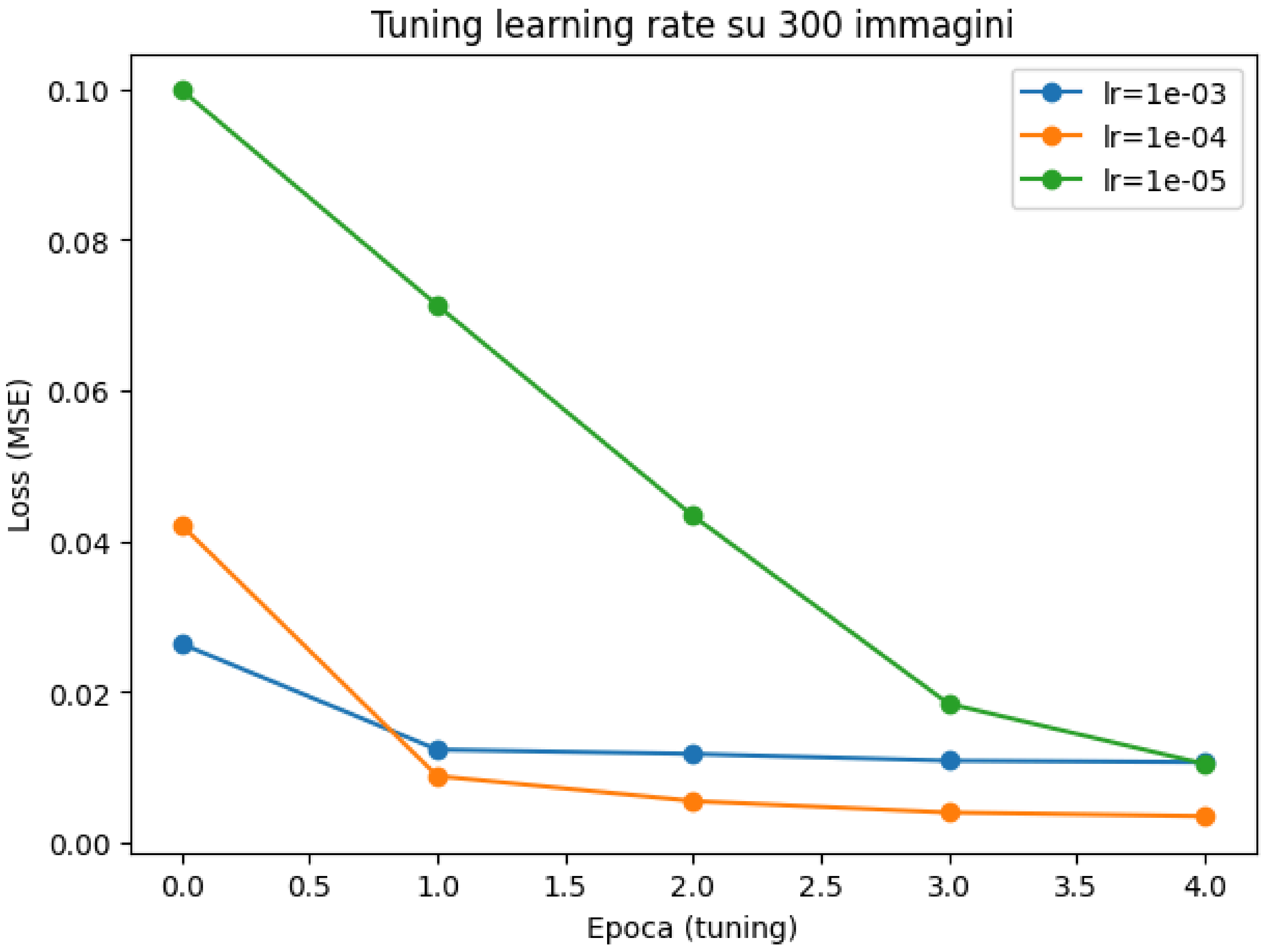


Fig. 8 — Learning rate tuning on 300 images, 5 epochs. 10^{-4} best trade-off: 10^{-3} plateaus, 10^{-5} too slow.

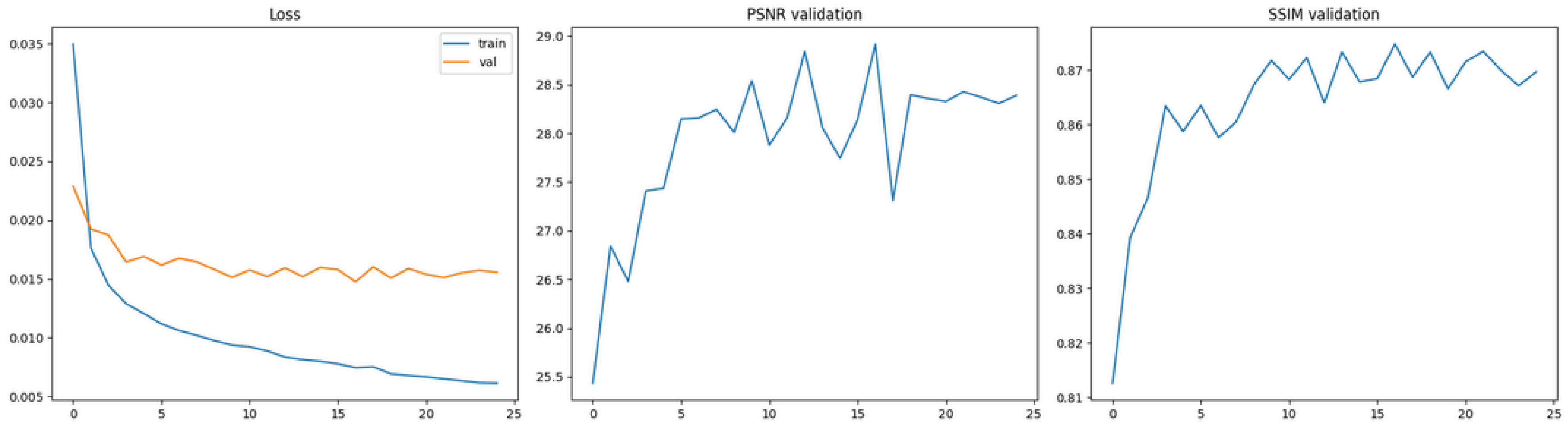


Fig. 9 — Train/val loss curves and validation PSNR/SSIM during full training (25 epochs), 15-angle configuration

Model selection

The final best model is chosen based on val_loss

UNet — example and mean metrics

90 angles	45 angles	30 angles	15 angles
23.35 dB ± 1.64	**23.32 dB** ± 1.48	**21.15 dB** ± 1.98	**20.91 dB** ± 2.12
SSIM **0.7189**	SSIM **0.7153**	SSIM **0.6250**	SSIM **0.6139**

Mean PSNR and SSIM (± std. dev.) over the entire test set (327 slices), UNet reconstruction only

Qualitative example (slice C081): ground truth vs UNet reconstruction across the 4 angular configurations, with PSNR/SSIM for the single case

GAN + DPS — phase 1: the generator

GENERATIVE

Two phases:
1) training an unconditional GAN generator, which learns only the distribution of plausible CT images, never seeing sinograms;
2) using the frozen generator as a prior in a data-guided sampling loop

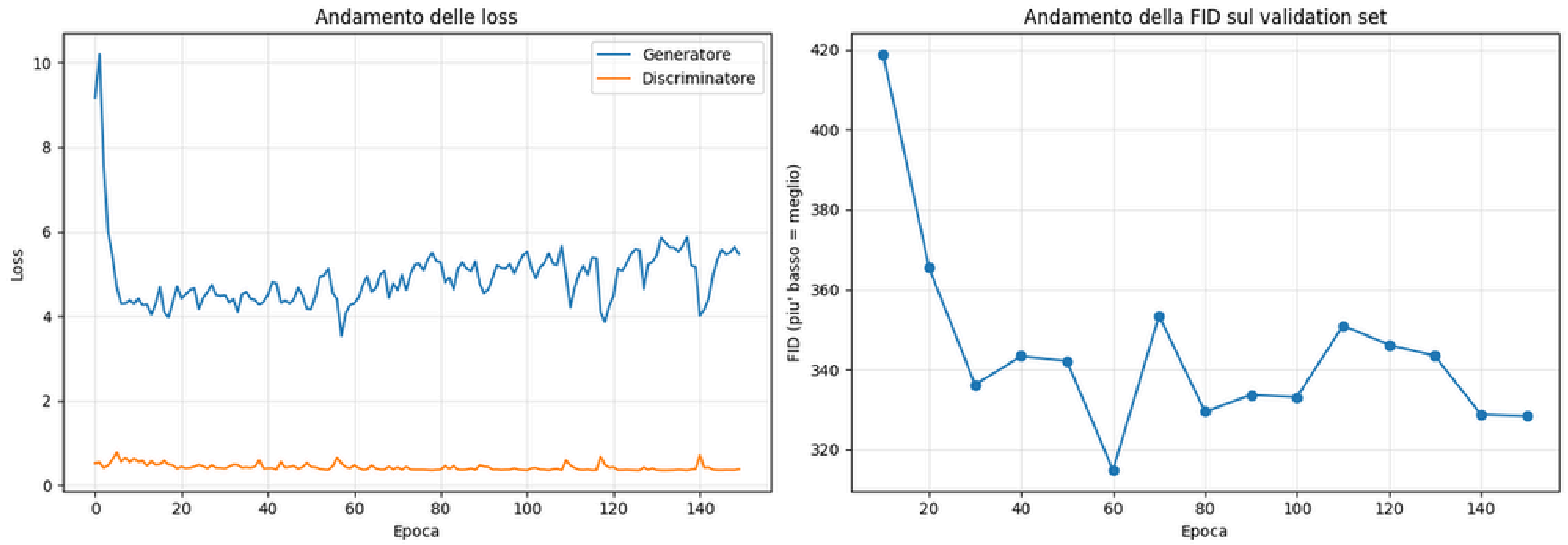


Fig. 10 — Loss (generator/discriminator) and FID on validation during 150 training epochs

- Generator: $z \in \mathbb{R}^{100}$ → 7 ConvTranspose2d layers → 256×256×1, channels 32·16→…→1, final sigmoid
- Discriminator: 6 Conv2d layers stride 2, BatchNorm + LeakyReLU, BCEWithLogitsLoss

150 epochs, batch 32, Adam betas=(0.5, 0.999), label smoothing (0.9)

Best-FID checkpointing: the absolute minimum is at epoch 60 (FID 314.77), after which FID worsens and oscillates up to 328.32 at epoch 150.

GAN + DPS — phase 2: guided sampling

DPS formulation (Diffusion Posterior Sampling adapted to a GAN prior)

$$L(z) = w\_data \cdot \| K\, G\theta(z) - y\delta \|_2^2 + \| z \|_2^2$$

The hypothesis does not hold up

The decay does not resolve the plateau: probably is a structural generator/discriminator imbalance, not too large a step near an optimum. The DPS phase uses the generator without decay (FID 314.77).

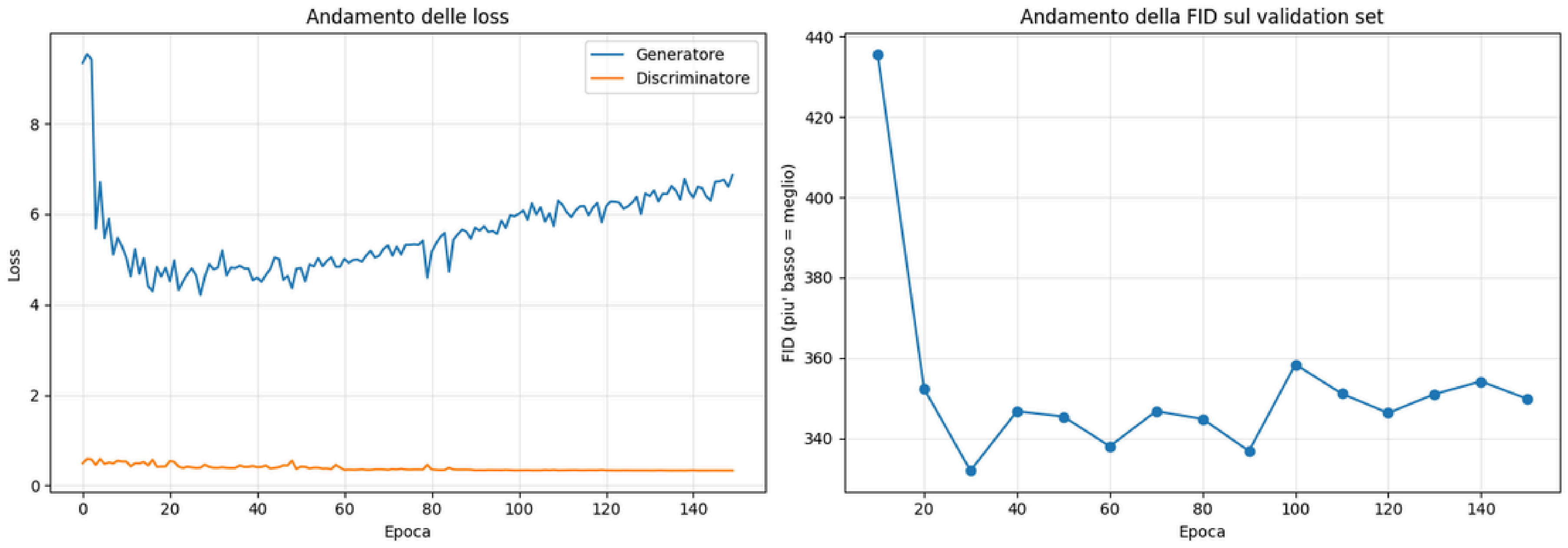


Fig. 11 — Attempt with learning-rate decay (halved every 30 epochs from epoch 60): the FID minimum (331.96, epoch 30) precedes the decay and is never reached again

GAN + DPS — tuning and structural limits

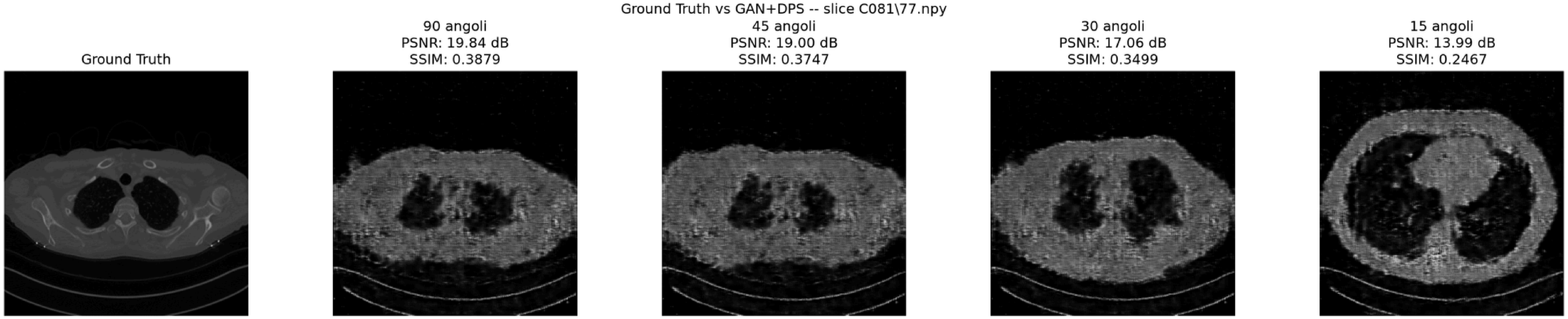
Grid search (12 validation samples, N_ITERS=300): hyperparameters chosen per configuration

Config.	data_weight	step_size	noise_scale	PSNR / SSIM
90	0.01	10^{-2}	0.00	17.44 dB / 0.327
45	0.01	10^{-2}	0.00	17.44 dB / 0.328
30	0.10	10^{-3}	0.02	16.41 dB / 0.307
15	0.10	10^{-3}	0.02	16.38 dB / 0.306

GAN + DPS — example and mean metrics

90 angles	45 angles	30 angles	15 angles
16.05 dB ± 1.43	**15.97 dB** ± 1.40	**15.48 dB** ± 1.29	**15.30 dB** ± 1.35
SSIM **0.2914**	SSIM **0.2893**	SSIM **0.2845**	SSIM **0.2799**

Mean PSNR and SSIM (± std. dev.) over the entire test set (327 slices), GAN+DPS reconstruction only



Qualitative example (slice C081): ground truth vs GAN+DPS reconstruction across the 4 angular configurations, with PSNR/SSIM for the single case

Weighted Total Variation

HYBRID

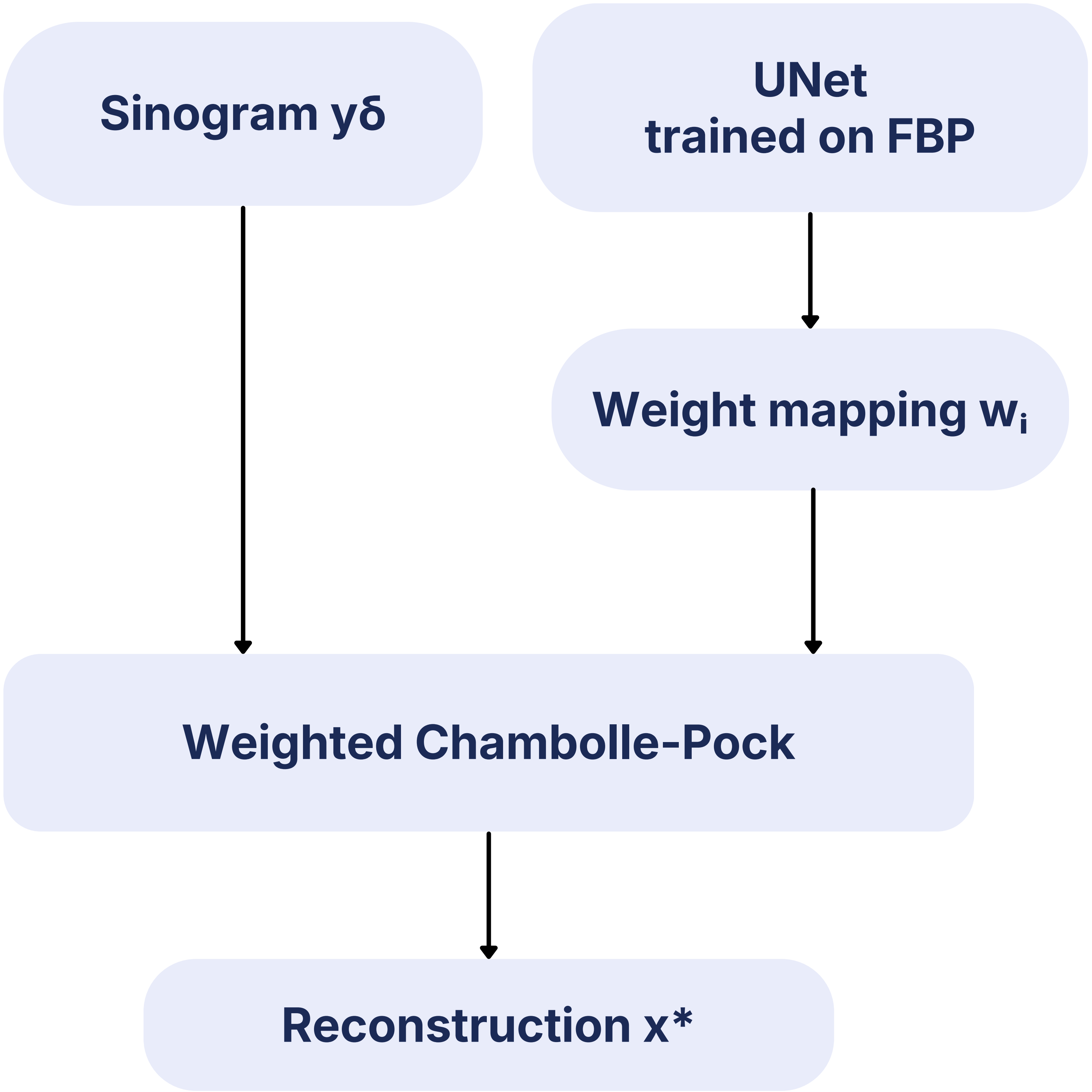
Replaces the global scalar weight λ of TV with a per-pixel weight wi:

$$TV\_w(x) = \Sigma i \;\; wi \, |Dx|i$$

wi computed from the intermediate image x̃ (output of the already-trained UNet, applied to the FBP):

$$wi = (\eta / \sqrt{}(\eta^2 + |D\tilde{x}|^2 i))\text{^}(1-p)$$

- Flat regions (gradient ≈ 0) → weight ≈ 1: full regularization
- Edges and details (high gradient) → weight ≈ 0: relaxed regularization



Weighted TV — tuning and limits

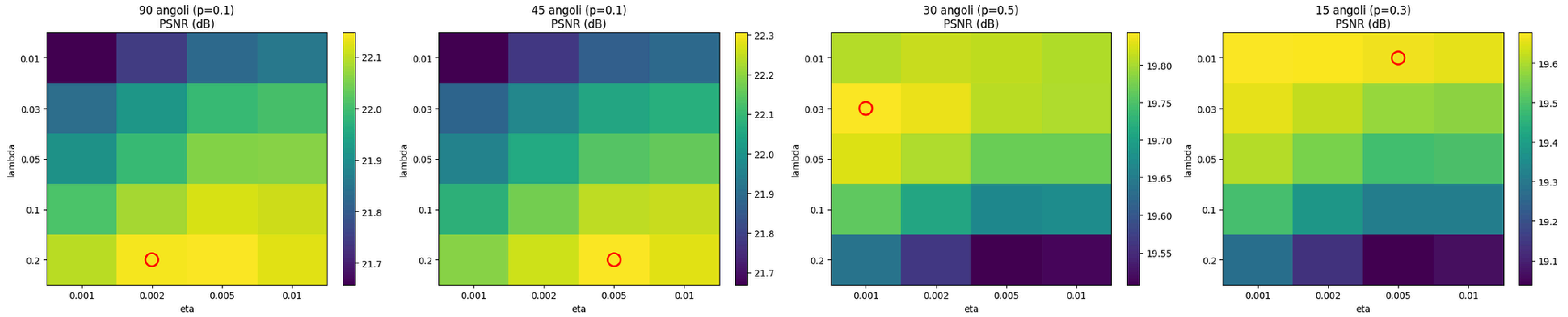


Fig. 12 — Heatmap of mean PSNR as λ and η vary (at the best p), per configuration; the circle marks the chosen combination

Config.	λ	η	p
90	0.20	$2\cdot10^{-3}$	0.1
45	0.20	$5\cdot10^{-3}$	0.1
30	0.03	$1\cdot10^{-3}$	0.5
15	0.01	$5\cdot10^{-3}$	0.3

Configurations with more angles (90, 45) → **low p** (strong contrast between weights); configurations with fewer angles (30, 15) → **high p** (softer contrast)

Dependence on **UNet**

No more efficient than TV

Weighted TV — example and mean metrics

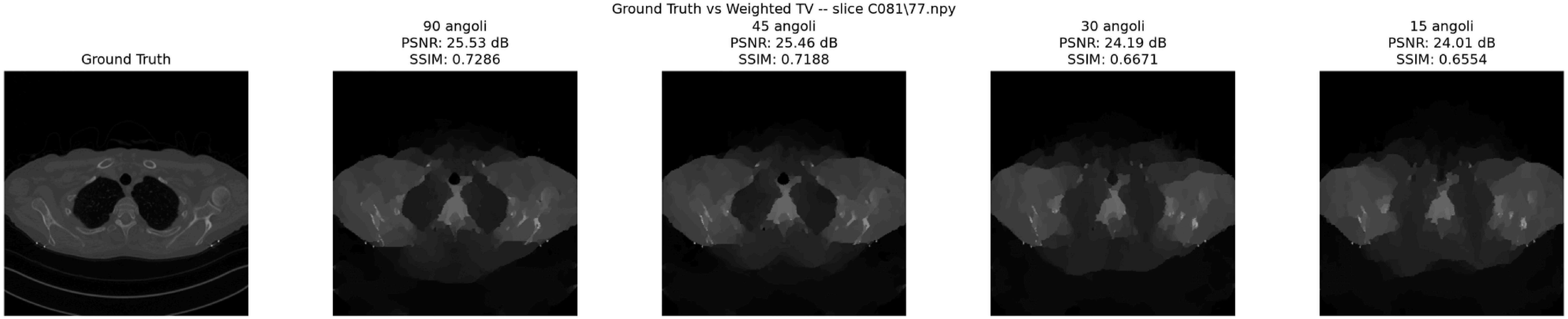
90 angles
24.06 dB ± 2.10
SSIM **0.7376**

45 angles
23.94 dB ± 2.07
SSIM **0.7309**

30 angles
21.91 dB ± 2.36
SSIM **0.6506**

15 angles
21.62 dB ± 2.27
SSIM **0.6299**

Mean PSNR and SSIM (± std. dev.) over the entire test set (327 slices), Weighted TV reconstruction only



Qualitative example (slice C081): ground truth vs Weighted TV reconstruction across the 4 angular configurations, with PSNR/SSIM for the single case

Quantitative comparison

Mean PSNR and SSIM (± std. dev.) over 327 slices of patient C081, versus ground truth

Method	Metric	90 angles	45 angles	30 angles	15 angles
TV	PSNR (dB) SSIM	23.83 ± 2.12 0.7257	23.79 ± 2.14 0.7221	21.75 ± 2.35 0.6426	21.51 ± 2.37 0.6231
UNet	PSNR (dB) SSIM	23.35 ± 1.64 0.7189	23.32 ± 1.48 0.7153	21.15 ± 1.98 0.6250	20.91 ± 2.12 0.6139
GAN+DPS	PSNR (dB) SSIM	16.05 ± 1.43 0.2914	15.97 ± 1.40 0.2893	15.48 ± 1.29 0.2845	15.30 ± 1.35 0.2799
Weighted TV	**PSNR (dB) SSIM**	**24.06 ± 2.10 0.7376**	**23.94 ± 2.07 0.7309**	**21.91 ± 2.36 0.6506**	**21.62 ± 2.27 0.6299**

Qualitative comparison

Same slice (patient C081), all angular configurations, all methods

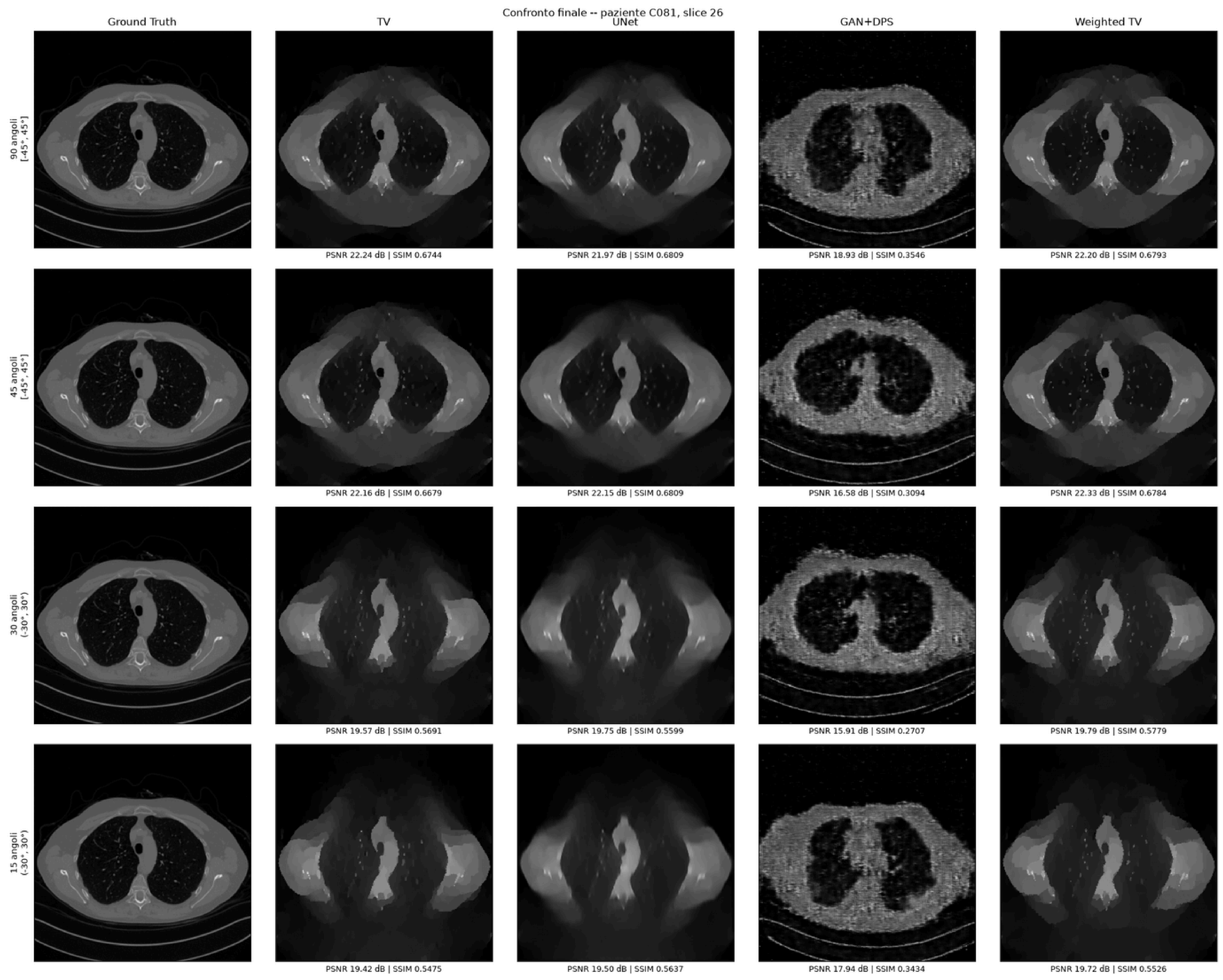


Fig. 14 — Overall qualitative comparison

UNet vs TV comparison

UNet's target is the TV reconstruction → UNet ≤ TV

The data confirm exactly this: UNet is very close to TV (–0.2/–0.6 dB relative to GT).

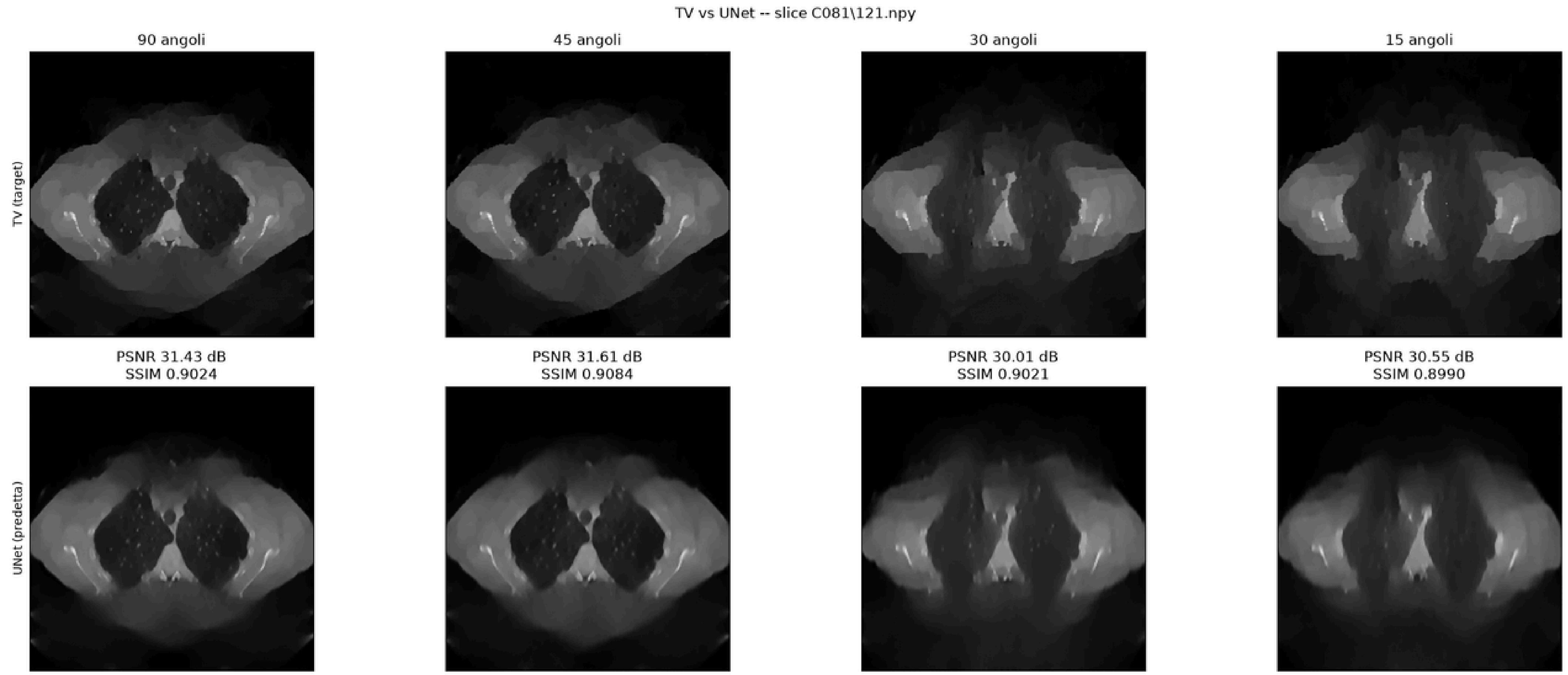


Fig. 13 — TV reconstruction (training target) vs UNet output, same slice, all configurations: the two reconstructions are visually almost indistinguishable (SSIM > 0.89)

Overall comparison between methods

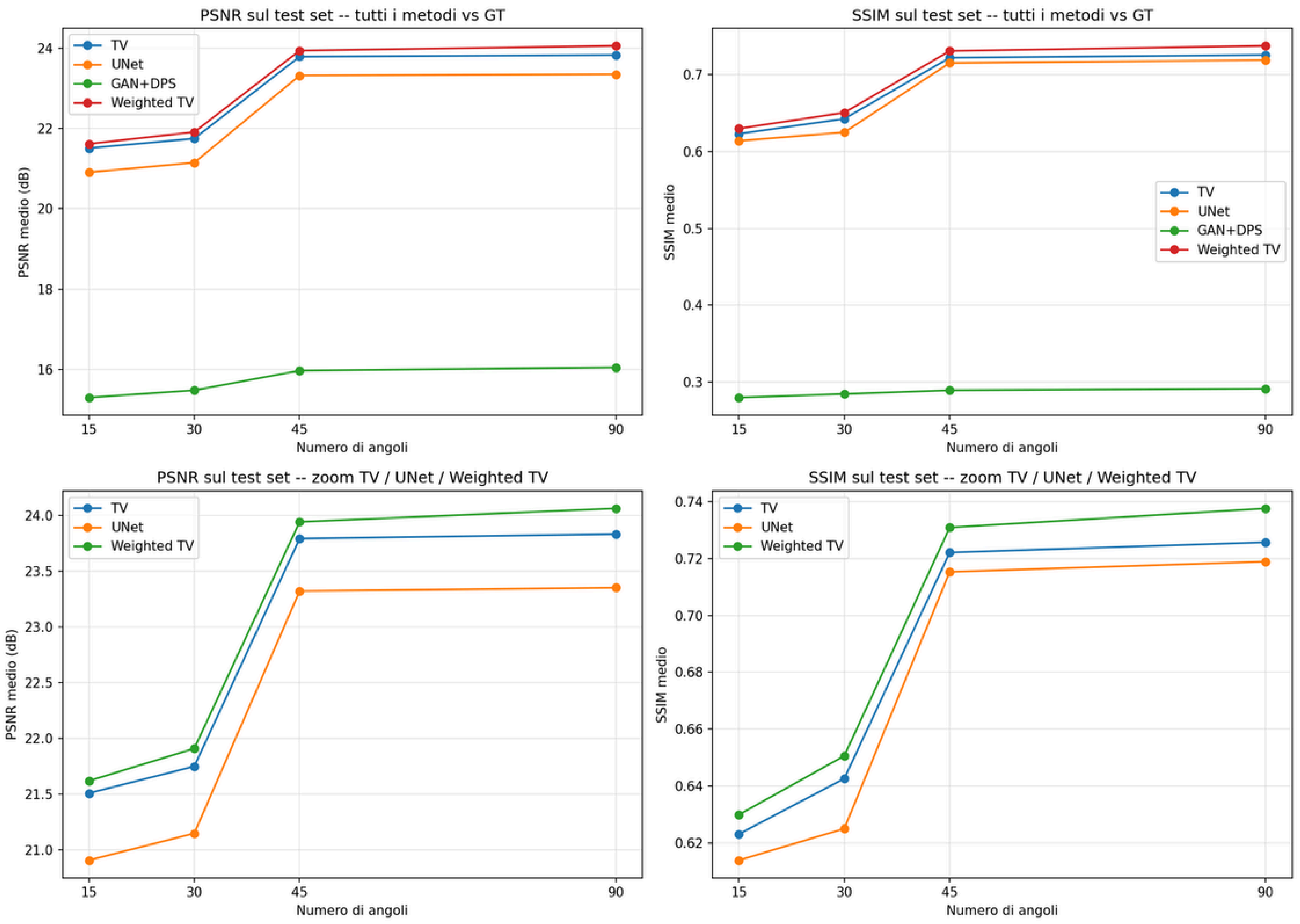


Fig. 15 — Mean PSNR/SSIM as the number of angles varies, all methods (top) and zoom on TV/UNet/Weighted TV (bottom)

Overall comparison between methods

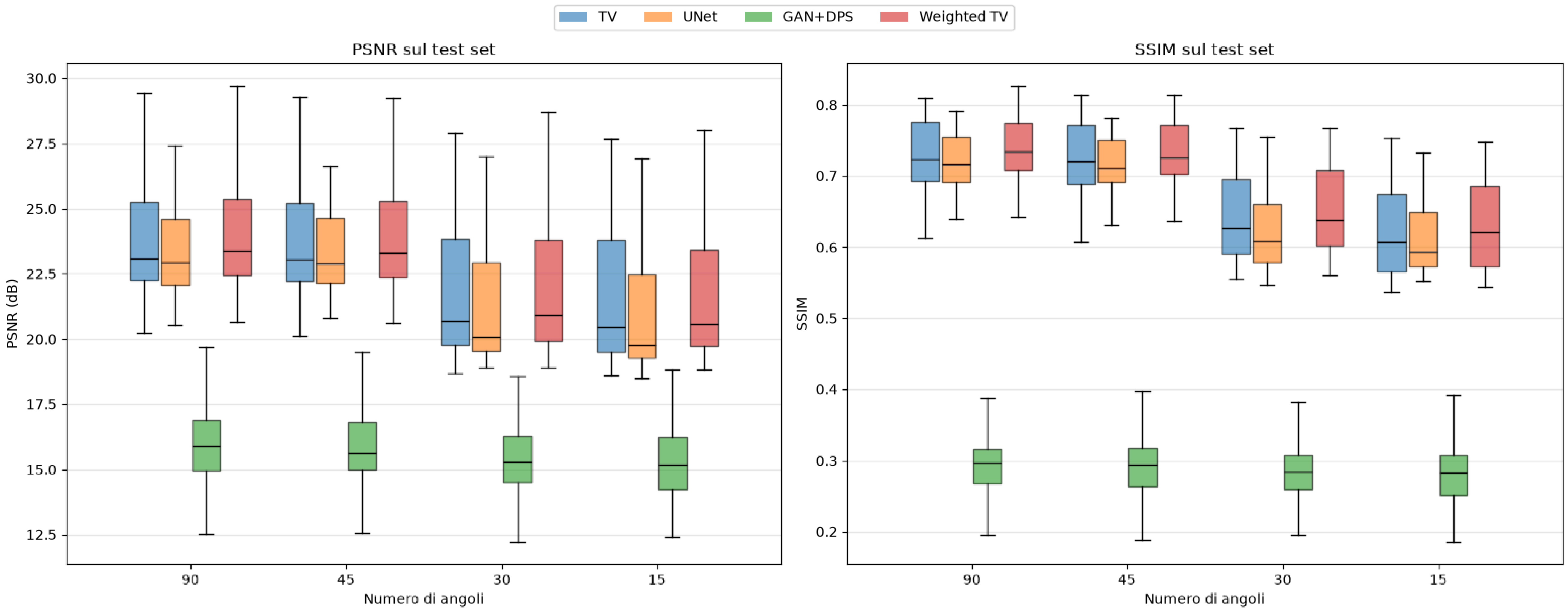


Fig. 16 — Distribution of PSNR and SSIM over the 327 test-set slices: GAN+DPS occupies a band completely separate from the other three methods, with no overlap

Conclusions

Stable ranking across all angular configurations:

Weighted TV ≥ **TV** ≥ **UNet** ≫ **GAN+DPS**

- Systematic gain of Weighted TV across all 4 configurations
- A learned prior improves a classical regularization if it guides its spatial structure, not if it replaces it entirely
- UNet remains a fast and effective approximation of TV (fidelity > 0.88 SSIM)
- GAN+DPS does not exploit its own theoretical advantage (generator trained on GT): the quality of the prior (FID) remains the dominant constraint

Limitations of the study

- Test set on a single patient
- TV/Weighted TV maxiter fixed at 300 due to time and computation constraints
- GAN+DPS complete but clearly weaker: the limitation seems to depend on the capacity of the generator and the dataset

THANK YOU FOR YOUR ATTENTION

Questions?